Symbolic signed-graph reduction for pencil relations

Mathematical implementation and development checks

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The existing pencil cohort has 543 verified candidate targets out of 1,260. This packet adds a native algebraic component, with no new cohort or performance measurement. Its candidate/rho cost ratio is unknown.

Algebraic interface

An input row is $\sum a_i X_i + \sum b_j S_j = 0$ over a prime field. At most two large-variable terms survive canonicalization; the S_j remain symbolic. The forest stores $X_i = s_i X_{\mathrm{parent}(i)} + u_i(S)$ and uses union by size. Root substitution leaves zero, one or two free roots. Zero emits a constraint, one anchors a root, and two join components by an invertible pivot. An unbalanced signed cycle anchors its root; it does not immediately yield an additional relation involving only small variables.

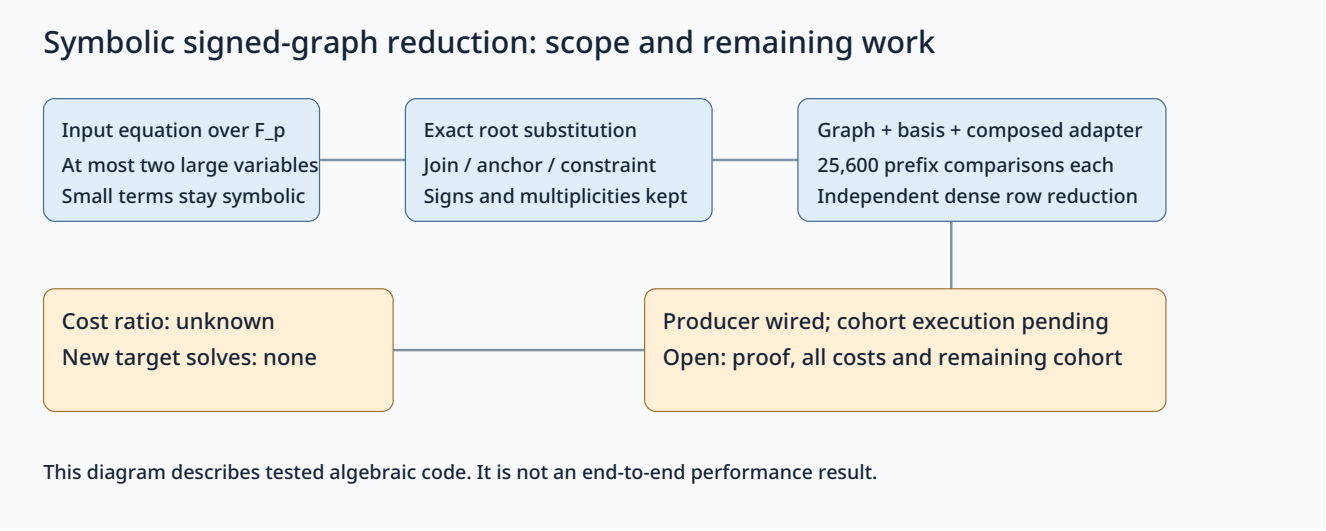
Recorded development checks

The Rust suite compares the emitted small-row space with independent dense row reduction after every prefix: 40 prefixes, 128 deterministic streams and five prime fields (2, 3, 5, 7 and 101), totaling 25,600 checks. Additional cases cover signed cycles, duplicate terms, rejected inputs, composites and a prime near the 64-bit limit. The test log is retained. The four Lean declarations cover pivot equivalence, substitution, balanced cycles and unbalanced cycles. Their direct compiler result and axiom report are retained separately. They do not formalize the full forest invariant or prove correspondence with Rust. The retained residual basis is now also compared with the independent reference at all 25,600 prefixes. Fourteen default library tests pass. Explicit working and retained coefficient caps leave the basis unchanged on rejection; callers must stop or retain that rejected row. Original residual diagnostics count failed attempts; the separate semantic and allocation observations described below extend their coverage. These are coefficient limits, not process-memory guarantees or a complete cost model. The composed adapter binds both stages to one field and fixed column partition, rejects three-large-variable rows explicitly, and retains an unincorporated constraint on terminal cap. Its row space is also checked at all 25,600 prefixes. The optional producer now connects this adapter to the existing geometric checks and matched control. Producer tests check signed coefficient parity, rejection of constant-only inconsistencies and phase attribution of symbolic events. Setup is a separate recorded phase. Semantic arithmetic counts include inversion multiplications and distinguish primality powers from inversions. Direct pivot lookup replaces full basis scans; no performance measurement is inferred. Residual diagnostics overlap this ledger and must not be double-charged. The producer has not run a cohort. Its new schemas explicitly mark costs incomplete and ratios null; a frozen partition and full accounting remain open. Allocation observations now cover cold setup, phases, the whole candidate envelope and rho. A grow/shrink regression checks payload preservation and requested-layout accounting. These bytes are not RSS or internal copying; rho's absolute peak includes still-live candidate and shared input state. The outer envelope includes between-phase work without losing peaks when inner phases reset. No runtime or memory comparison has been measured. Sparse maps now record API calls, ordered comparisons, yielded entries and cloned

entries. Exact iterator size hints and numeric ordering are preserved; JSON serialization is unmetered. These are semantic events, not all CPU work. Counts depend on the recorded standard-library version. The current suite has 14 default library tests, 15 with the producer feature and five adapter tests. The producer now verifies field, native arithmetic and symbolic phase sums, and retains interval overhead separately. Scalar inverse instrumentation was asymmetric: pencil counts inner multiplications; the reference records one inverse. A native normalization removes pencil inverse multiplications for a common atomic scalar-I model, tested against both actual inverse implementations. The model remains a draft with unfrozen weights; no historical ratio is revised.

Next integration gates

Requirement	Status
Prime-field row kernel and independent prefix tests	Implemented and tested
Forest-state induction and Rust correspondence	Open
Geometric producer wiring	Compile-tested only
Fixed small-base selection and admission accounting	Open
Full symbolic-vector, memory and residual elimination costs	Unmeasured
Remaining 717 frozen candidate targets	No new results
End-to-end cost against matched rho	Unknown

Sources and limitations

The graph construction follows the mathematical distinction between full relations and cycles leaving a large prime in Gaudry, Theriault and Thome, double large prime variation, sections 2.3 and 3.1 (2005 version). Its genus-at-least-three complexity result is not asserted for this genus-2 pencil stream. Existing repository graph and large-prime eliminators informed the implementation. The editable graph is `reduction.svg`; the machine-readable check table is `checks.csv`. All results are development checks; no runtime, memory improvement or full-cohort completion is inferred from them.